MA 765: Topics in Non-linear Algebra

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1 Gröbner Basis

Denote by $\leq$ the coordinate wise partial order on $\mathbb{N}_0^n$. $(a_1, \dots, a_n) \leq (b_1, \dots, b_n)$ if $a_i \leq b_i$ for $i \in [n]$.
Divisibility is a partial order on monomials.

Theorem 1.1 (Dicksen's lemma)**.** *Every infinite subset of $\mathbb{N}_0^n$ contains elements a, b with $a < b$.*

Proof. The proof follows by induction. Let $\mathcal{M} \subset \mathbb{N}_0^n$ be an infinite subset. For any $i \in \mathbb{N}_0$ define $\mathcal{M}$,

$$\mathcal{M}_i = \{a = (a_1, \dots, a_n) \in \mathbb{N}_0^{n-1} \ : \ (a, i) \in \mathcal{M}\}$$

If $\mathcal{M}_i$ is finite, then look at $\bigcup_i \in \mathbb{N}_0$, which has finitely many and atleast one minimal element. Thus there is some $j \in \mathbb{N}$ such that $\bigcup_{i=0}^{j} \mathcal{M}_i$ with $a \in \mathcal{M}_i$ for some $i \leq j$.Hence $(a, i) < (b, k)$.

□

Corollary 1.2. *For any $\phi \neq \mathcal{M} \subset \mathbb{N}_0^n$ the set of minimal elemnents wrt $<$ is nonempty and finite.*

1.3. Monomial order on $\mathbb{N}_0^n$ is a relation $\prec$ such that

1. If $a \neq b$ then $\prec b$ or $b \prec a$.
2. If $A \prec b$ and $b \prec c$ then $a \prec c$.
3. $(0, \dots, 0) \prec a$ for any $a \in \mathbb{N}_0^n$.
4. $a \prec b$ then $a + c \prec b + c$ for any $c \in \mathbb{N}_0^n$.

First two conditions together imply that $\prec$ is a total order.

Remark 1.4.
1. *If $a < b$, then $a \prec b$, i.e., any monomial order refines coordinate wise partial order.*
2. *Any monomial order on $\mathbb{N}_0^n$ gives total order on monomials in $k[X_n]$*

Example 1.5. *Lexicographic order: Left most non zero entry of $a - b$ is positive the $a >_{lex} b$.*

degree lex order if either $|a| > |b|$ or $|a| = |b|$ and right most entry of $a - b$ is negative.

degree reverse lex order, same as above but right most entry of $a - b$ is negative.

Proposition 1.6. *For any monomial order $<$ on $\mathbb{N}_0^n$ with $\phi \neq \mathcal{M} \subset \mathbb{N}_0^n$ has a unique minmial element.*

Proof. Dicksen's lemma gives that $\mathcal{M}$ has a finite nom empty set of minmal elements wrt coordinate wise order. The minmal elet of thse wrt $<$ is the desired elemnt. □

1.7. Fix a monomial order $<$ on $k[X_n]$

1. The initial monomial $\mathrm{im}_<(F)$ of $f = \sum c_a x^a$ is the largest monomial wrt $<$ appearing in f with non zero coefficients.

2. The leading term $\mathrm{lt}_<(f) := c_a x^a$ with $x^a = \mathrm{im}(f)$
3. The initial ideal of an ideal I is $\mathrm{im}(I) = \langle \mathrm{im}(f) : f \in I \rangle$.

If G is a generating set for an ideal I then $\langle \mathrm{im}(g) : g \in G \rangle \subset \mathrm{im}(I)$ and this inclusion can be strict.

Proposition 1.8. *Let $<$ be a monomial order on $k[X_n]$. Every ideal I has a finite subest $\mathcal{G}$ such that $\mathrm{im}(I) = \langle \mathrm{im}(g) : g \in \mathcal{G} \rangle$.*
Any such $\mathcal{G}$ is called a Gröbner Basis of I wrt $<$.

Proof. The set of monomials in $\mathrm{im}(I)$ has a finite and nonempty subset of minmal elements wrt divisibility, say $m_1, \dots, m_s$. Thus $\mathrm{im}(I) = \langle m_1, \dots, m_s \rangle$. Every monomial in $\mathrm{im}(I)$ is the initial monomial of some $f \in I$. hence there exists $f_1 \dots, f_s \in I$ with $\mathrm{im}(f_i) = m_i$. Thus $\{f_i\}$ is a Gröbner basis. □

Theorem 1.9. *If $\mathcal{G}$ is a Gröbner basis of I, then $I = \langle \mathcal{G} \rangle$*

Note that Hilbert's basis theorem is a simple corollary of this.

Proof. We argue by contradiction. By 1.6, choose $f \in I - \langle \mathcal{G} \rangle$ such that $\mathrm{im}(f)$ is minimal. Call $\mathrm{im}(f) = x^b$. $x^b \in \mathrm{im}(I) = \langle \mathrm{im}(g) : g \in \mathcal{G} \rangle$. There exists $g \in \mathcal{G}$ such that $\mathrm{im}(g) | x^b$, say $x^b = x^c \cdot \mathrm{im}(g)$.
$\mathrm{im}(f = x^c \lambda g) < x^b = \mathrm{im}(f)$ where $\lambda = \mathrm{lt}(f)/x^c\,\mathrm{lt}(g) \in k$. But $f - x^c \lambda g \in I - \langle \mathcal{G} \rangle$, which is a contradiction to the minimality. □

Lemma 1.10. *Consider $f, g_1, \dots, g_s \in k[X_n]$, with $g_i \neq 0$. Then for any minmal order $<$, there exists $q_1 \dots, q_s, r \in k[X_n]$ such that*

1. $f + \sum_{i=1}^{s} q_i g_i + r$
2. $\mathrm{im}(f) \geq \mathrm{im}(q_i g_i) \forall i$ *(note that* $\mathrm{im}(f) = \mathrm{im}(q_i g_i)$ *for some* i*)*
3. $\mathrm{im}(r)$ *is not divisible by* $\mathrm{im}(g_i)$ *for any* i*.*

We say f reduces to r by $\{g_1, \dots, g_s\}$.

1.11 Division Algorithm. Input: $f, g_1 \dots g_s$
Output: $q_1, \dots q_s, r$ satisfying the properties 1-3.

1. Set $r = 0$, $p = gf, q_1 = \cdots = q_s = 0$
2. While $p \neq 0$ do :

   If $\mathrm{im}(g_i)$ divides $\mathrm{im}(p)$ for some $i \in [s]$, then set $q_i = q_i + \frac{lt(p)}{lt(g_i)}$ and $p_i = p - \frac{lt(p)}{lt(g_i)} g_i$
   else set $r = r + \mathrm{lt}(p)$, $p = p - \mathrm{lt}(p)$
3. Return $q_1, \dots, q_s, r$

Example 1.12. *Consider Lexicographic ordering with $x > y$ and $f = x^2y + xy^2 + y^2$ and $g_1 = xy - 1,\ g_2 = y^2 - 1$.*

p	$xy-1$	y^2-1	r
$x^2y + xy^2 + y^2 - x^2y + x$	x		
$xy^2 + y^2 + x - xy^2 + y$	y		
$y^2 + x + y - x$			x
$y^2 + y - y^2 + 1$	1		
$y + 1 - y$			y
1			1

Corollary 1.13. *Buchberger's criteria Let $\mathcal{G}$ be a finite subset of I. Then $\mathcal{G}$ is a Gröbner basis of I iff each $f \in I$ can be reduced to 0 by $\mathcal{G}$.*

Proof. If f reduces to r by $\mathcal{G}$, then $\text{im}(g_i)$ does not divide $\text{im}(r)$, for all i. Howevert $f - r \in \langle \mathcal{G} \rangle$ and $r \in I$ and $\mathcal{G}$ is a Gröbner basis of I. So there exists some $g \in \mathcal{G}$ such that $\text{im}(r)$ is divisible by $\text{im}(g)$, which forces $r = 0$.

Conversely, we have $f = \sum q_i g_i$ with $g_i \in \mathcal{G}$ and $\text{im}(q_i g_i) \leq \text{im}(f)$. Hence equality for some i and so $\text{im}(f) \in \langle \text{im}(g) : g \in \mathcal{G} \rangle$. □

Proposition 1.14. *Let $<$ be a monomial order. Then*

1. *Let B be the set of monomial in $k[X_n] - \text{im}(I)$. Then $\bar{B} \subset k[X_n]/I$ is a k vsp basis.*
2. *If $\mathcal{G}$ is a Gröbner basis of I, then the remainder of f by $\mathcal{G}$ is unique and does not depnd on the choice of $\mathcal{G}$.*

Proof. 1. If $p = \sum \lambda_i m_i \in I$ with $m_i \in B$, then $\text{im}(p) \in \text{im}(I)$, but $\text{im}(p) = \text{im}(m_i) \notin I$. To show $\bar{B}$ spans.$m \in k[X_n]$ such that $\bar{m} \notin$span$(\bar{B})$.

   Take $\min\{m\} = m$ where $m \notin B$. So we have $m \in \text{im}(I)$. There exists $f \in I$ such that $\text{im}(f) = m$. So any monomial in $f - \text{lt}(f) + I = f - \lambda m + I$ is in span $(\bar{B})$. So $\lambda m + I = p - f + I \in$span$(\bar{B})$. This leads to a contradiction

□

Definition 1.15. *1. For terms $\lambda x^a, \mu x^b \quad (\lambda, \mu \in k)$ denote by*

$$\gcd\left(\lambda x^a, \mu x^b\right) = \gcd\left(x^a, x^b\right)$$
$$\text{lcm}\left(\lambda x^a, \mu x^b\right) = \text{lcm}\left(x^a, x^b\right)$$

2. *For $0 \neq g,\ h \in k[x_n]$, their s -polynomial (wrt monomial order $<$)*

$$S(g,h) := \frac{\text{lt}(h)}{\gcd(\text{lt}(h), \text{lt}(g))} g = \frac{\text{lt}(g)}{\gcd(\text{lt}(h), \text{lt}(g))} h.$$

1.16. Buchberger's Algorithm for computing Gröbner basis Input: $f_1, \ldots, f_s \in k[X_n]$ in monomial order.
Output: Gröbner basis $\mathcal{G}$ of $I = \langle f_1, \ldots, f_s \rangle$ wrt $<$.

1. Set $\eta = \langle f_1 \cdots, f_s \rangle$
2. Order the elements of $\mathcal{G}$ as $f_1, \ldots, f_t$
3. For $1 \leq i < j \leq t$ do: Reduce $S(g_i, g_j)$ to r by g. If $r \neq 0$, then set $\mathcal{G} := \mathcal{G} \cup \{r\}$ and go to step 2 .
4. Return $\mathcal{G}$

Remark 1.17. *The algorithm computes a Gröbner basis. It terminate vecause in case* $r \neq 0. \operatorname{im}(r) \notin \langle \operatorname{im}(g_1), \cdots, \operatorname{im}(g_t) \rangle$.

1.18. Extension to submodules of finitely generated $k[X_n]$ module So $F = k[X_n]^r = \bigoplus_{i=1}^{r} k[X_n]e_i$.

Monomials in G are of the form $x^a e_i$ and terms are $\lambda x^a e_i$ with $\lambda \in k$.
A monomial order on FF is a total order on the monomials satisfying:
If $x^a \neq 1$, then $m_1 < m_2 \implies m_1 < x^a,_1 < x^a m_2$, for monomials m_i in F.

Given a monomial order on the polynomial ring $k[X_n]$ and an order on $\{e_i\}$. We obtain a monomial ordering on F by ordering $\mathbb{N}_0^n \times [r]$ or $[r] \times \mathbb{N}_0^r$ lexicographically.

Dicksen's lemma can be extended to monomials in F. We also define $\operatorname{im}, \operatorname{lt}$ with respect to $<$ analogously. There is also a division algorithm.

Theorem 1.19. *1. Every submodule M of F has finite Gröbner basis and the basis generates M.*

2. If B is the set of monomials in $F - \operatorname{im}(M)$, then $\bar{B} \subset F/M$ form a k basis of F/M.